Reranking

Finding the right page, then putting it first

Weekend 5 · Retrieval Augmented Generation

Carlos Cotrini

Section 1

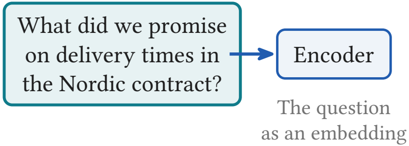
From retrieval to reranking

1

Where this morning left us

What did we promise on delivery times in the Nordic contract?

Where this morning left us



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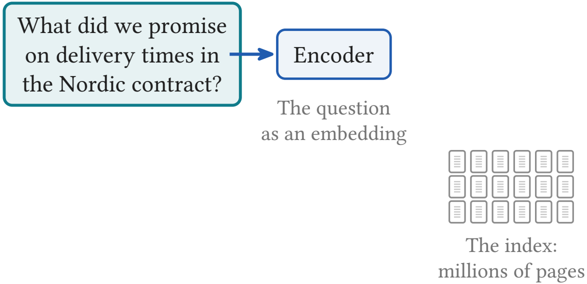

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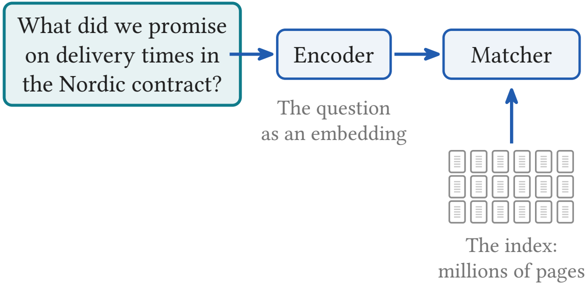

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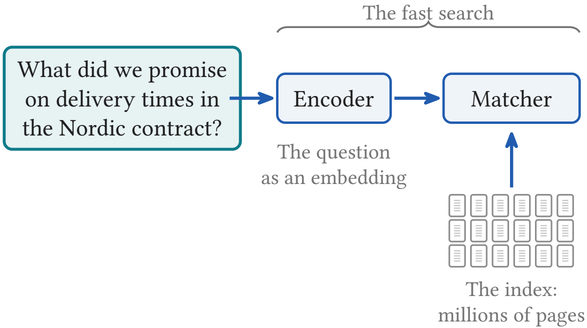

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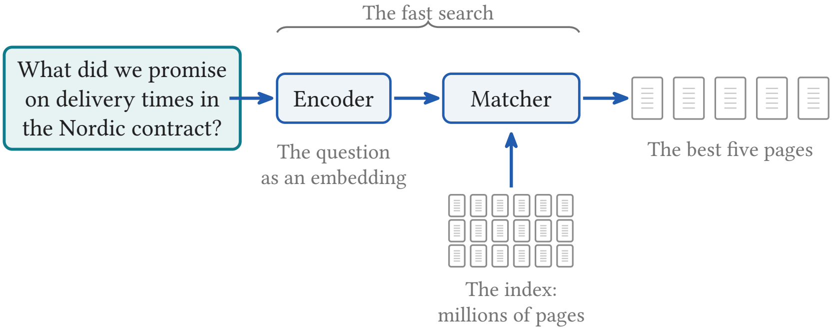

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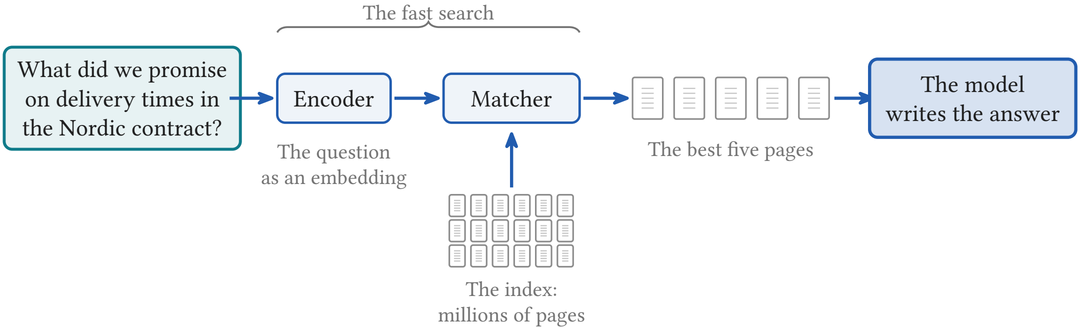
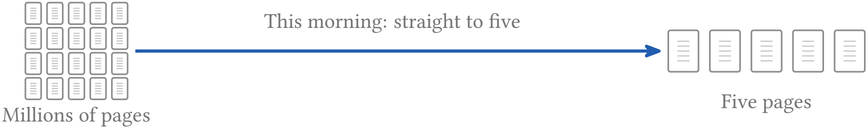


Reranking: choose from a larger shortlist



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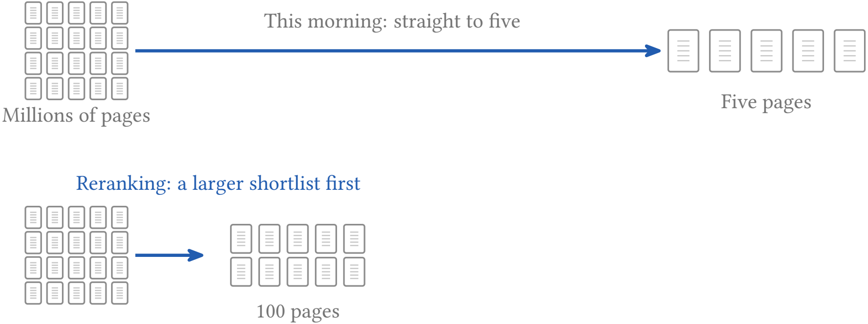

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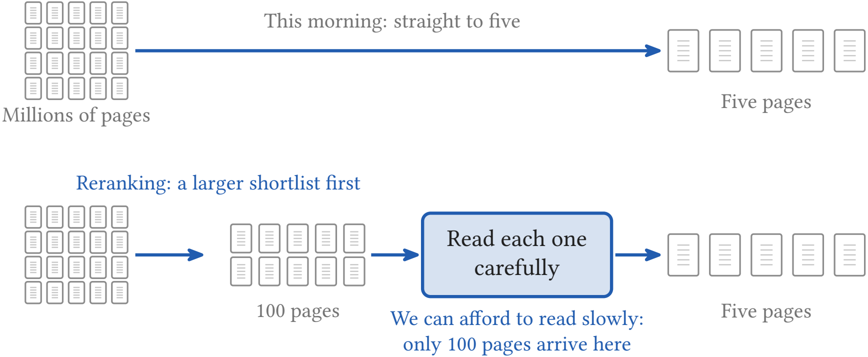
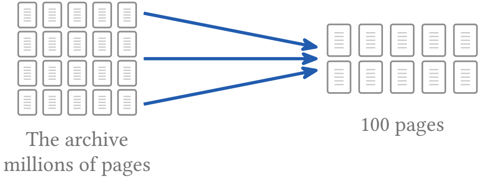


Two different jobs

The archive
millions of pages

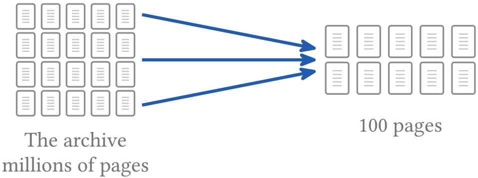
Two different jobs



Two different jobs

Find

Wide, cheap, skim everything



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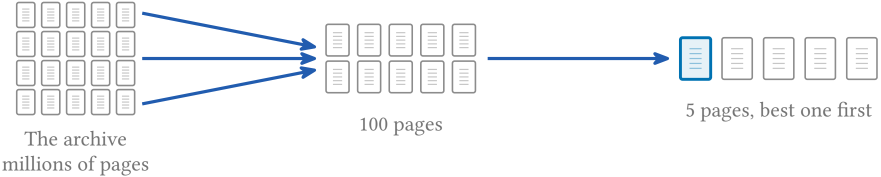

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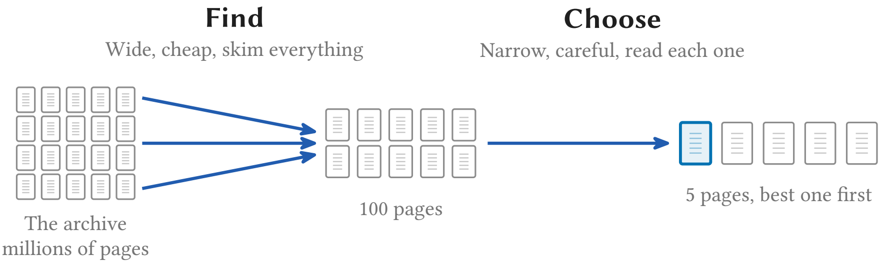


Section 2

Three families of reranker

2

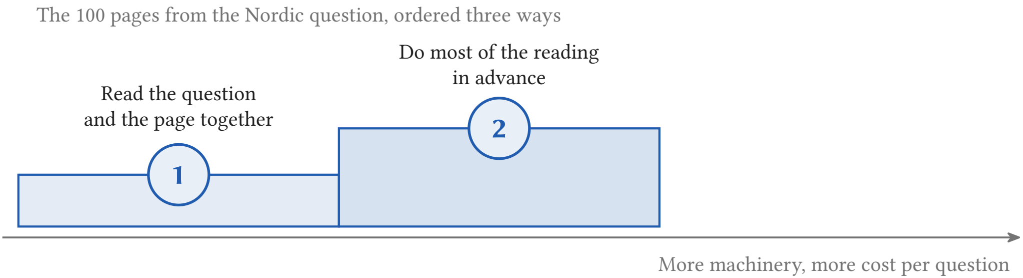
Three ways, cheapest first

The 100 pages from the Nordic question, ordered three ways

Read the question
and the page together

More machinery, more cost per question

Three ways, cheapest first



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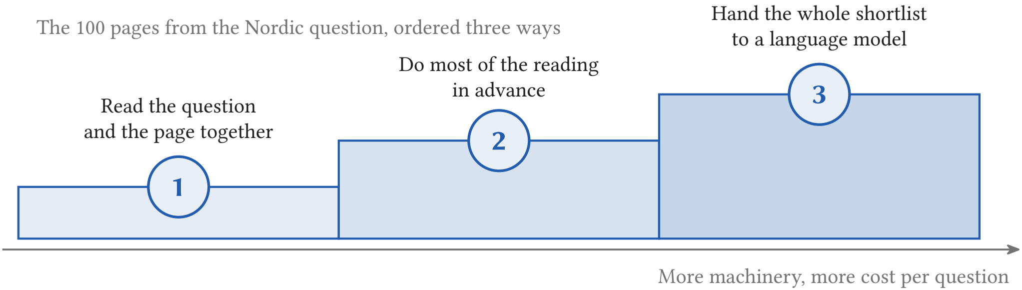


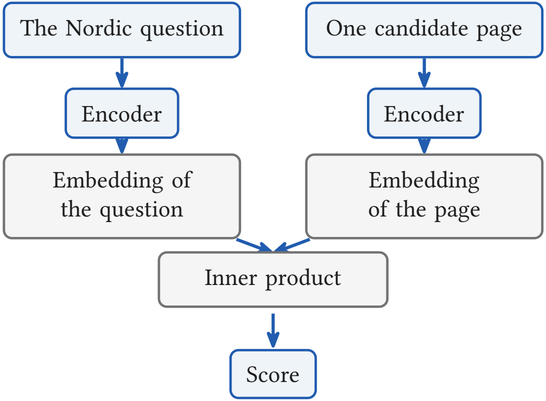
1

Rung 1 of three

Read the question and the page together

Read the question and the page together

This morning: two encoders, one inner product



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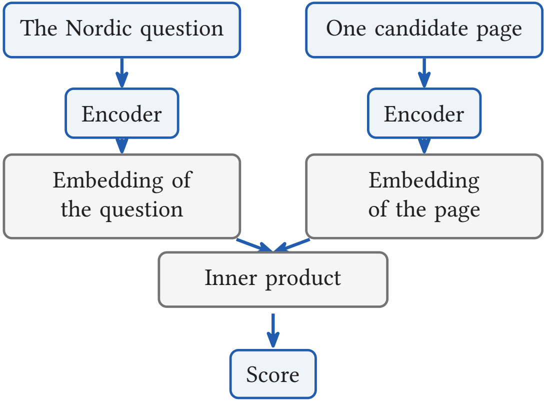

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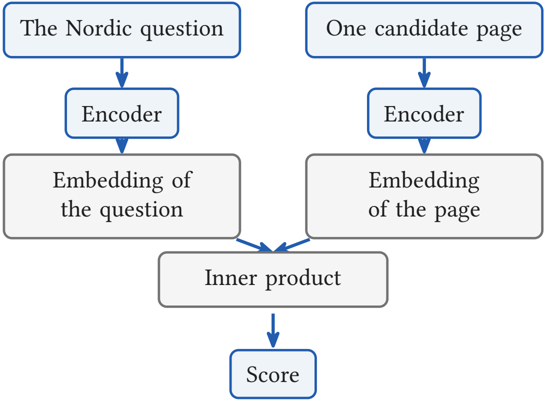

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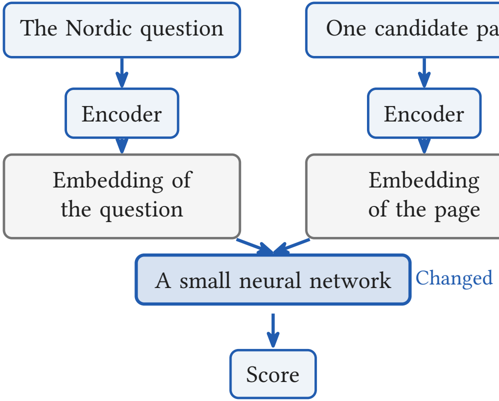

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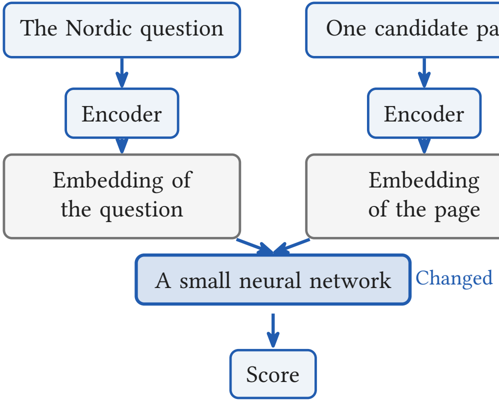

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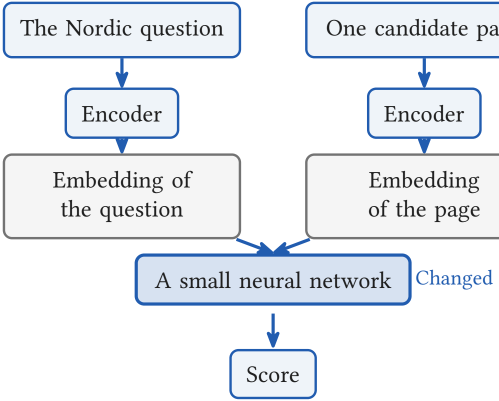

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Encoded together, one embedding can carry how the two relate, not only what each says alone.

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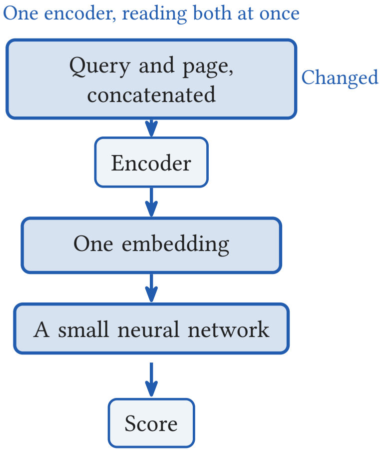

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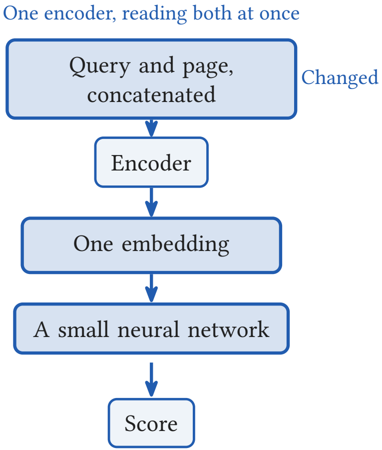

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Why encode the question and the page apart?

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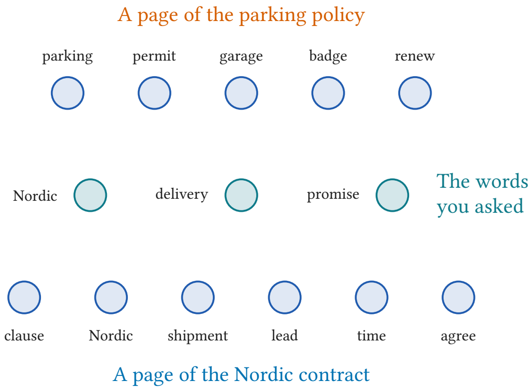
This is the cross encoder. One model run for every candidate page.

Rung 2 of three

2

Do most of the reading in advance

Best match, one question word at a time



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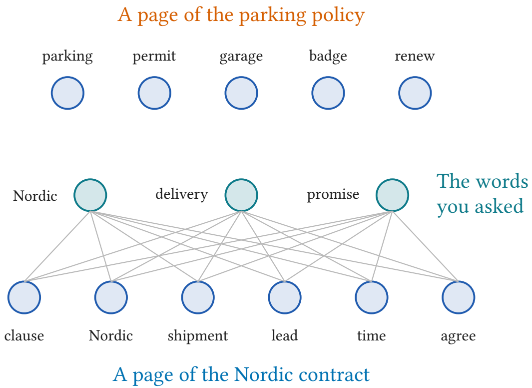

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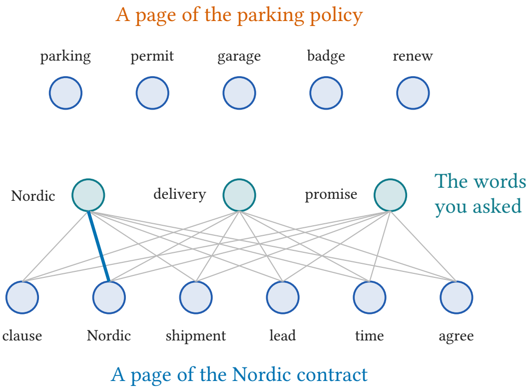
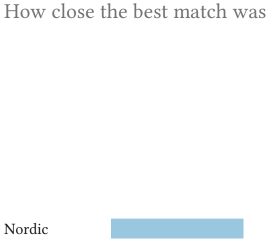
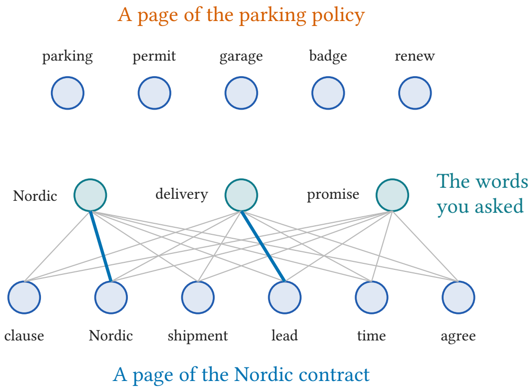




Illustrative, not measured.

Best match, one question word at a time



How close the best match was

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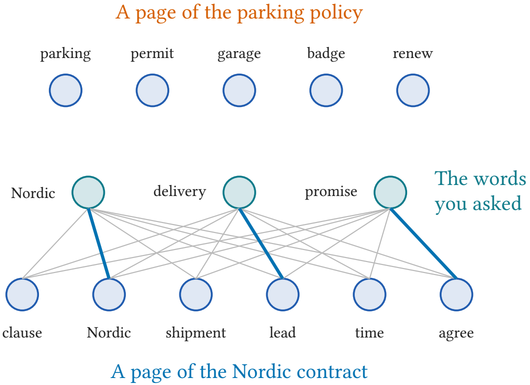

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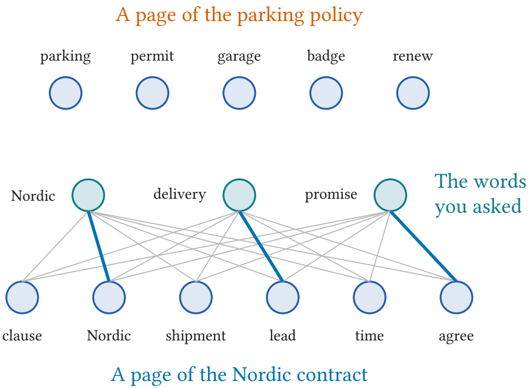

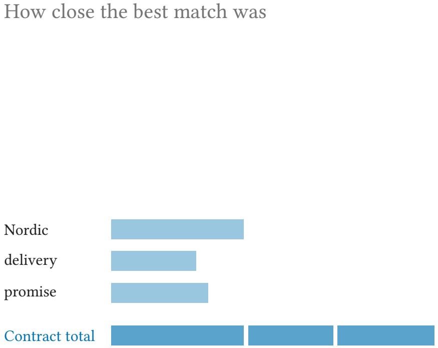

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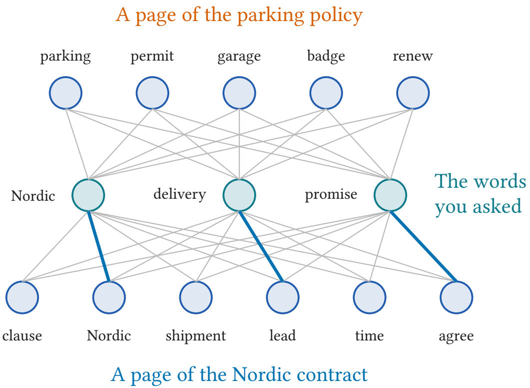

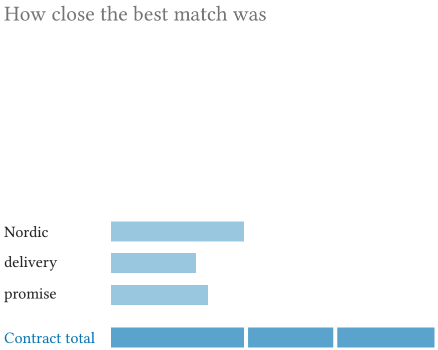

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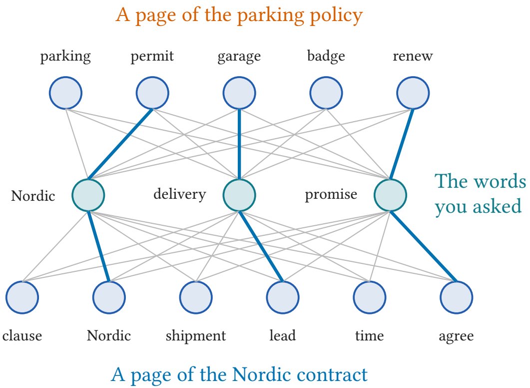

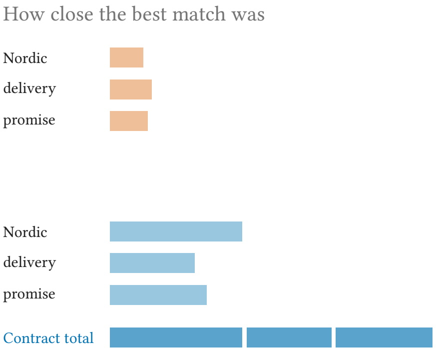

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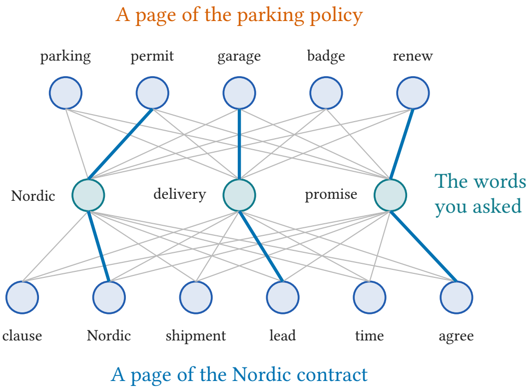
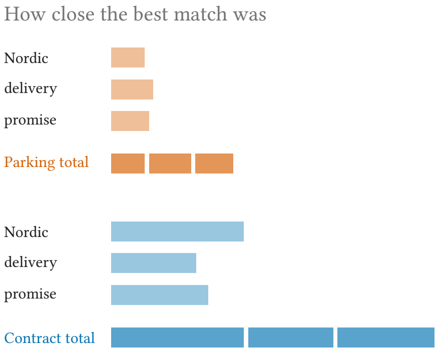




Illustrative, not measured.

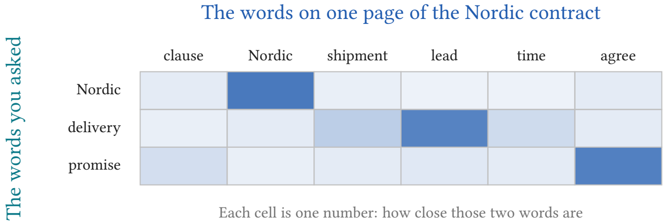
What the best match score actually computes

The words on one page of the Nordic contract

The words you asked

	clause	Nordic	shipment	lead	time	agree
Nordic						
delivery						
promise						

What the best match score actually computes



Each cell is one number: how close those two words are

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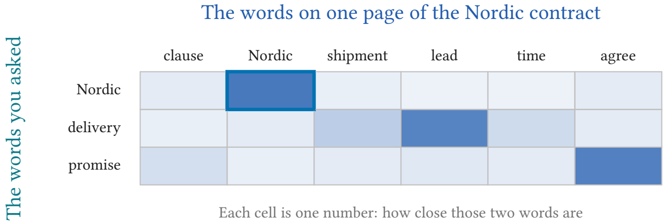

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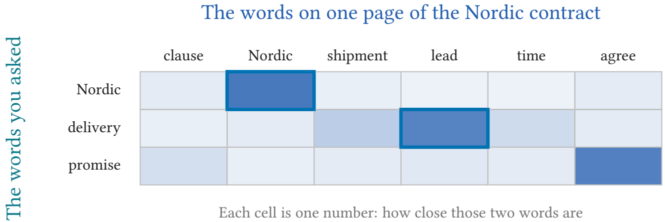

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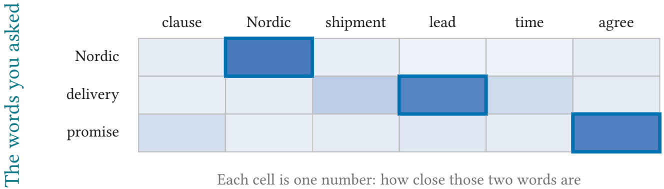

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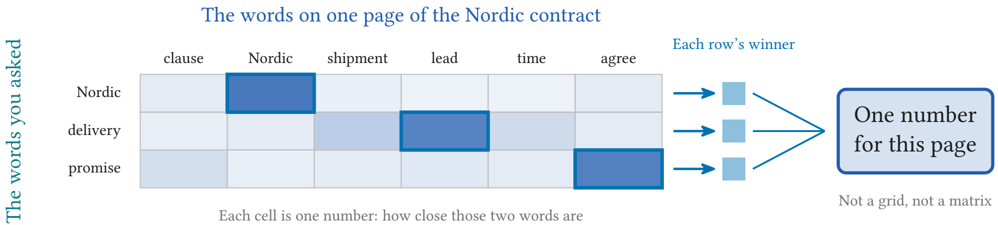
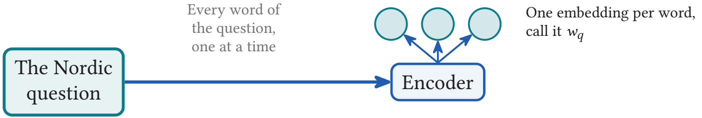


The best match score



ColBERT, SIGIR 2020.

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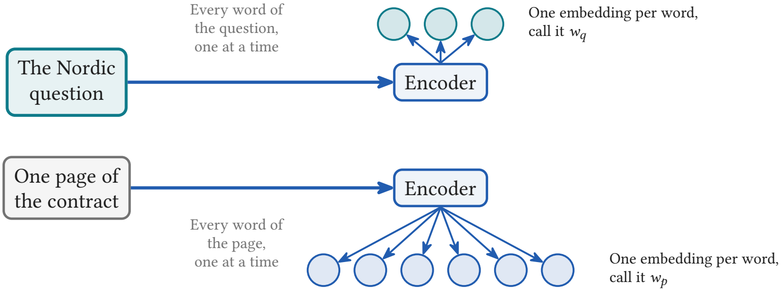

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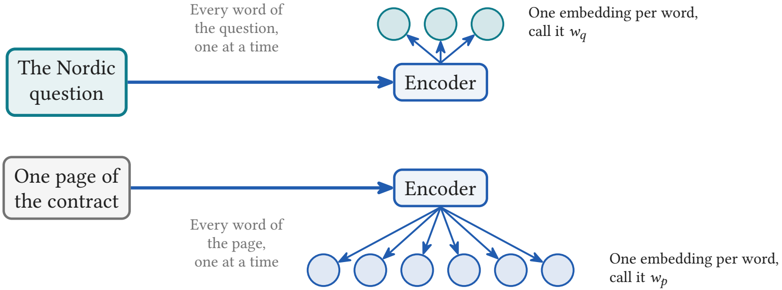
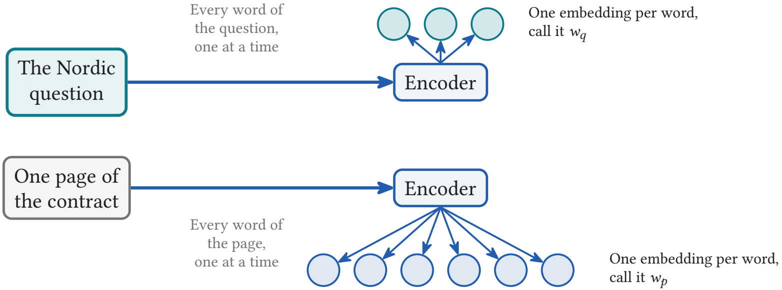


$$\underbrace{\text{match}(w_q, w_p)}_{\text{One question word against one page word}}$$

ColBERT, SIGIR 2020.

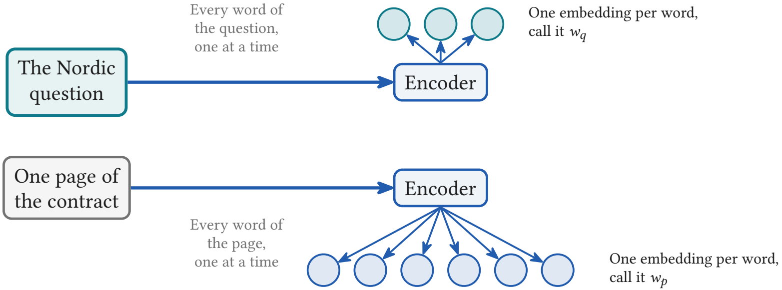
The best match score



$$\underbrace{\max_{\text{page words } w_p}}_{\text{Keep only its best match}} \quad \text{match}(w_q, w_p)$$

ColBERT, SIGIR 2020.

The best match score



$$\text{score} = \underbrace{\sum_{\text{question words } w_q}}_{\text{Every word you asked}} \underbrace{\max_{\text{page words } w_p}}_{\text{Its best match}} \text{match}(w_q, w_p)$$

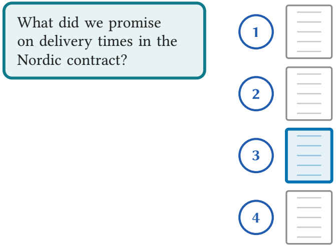
ColBERT, SIGIR 2020.

3

Rung 3 of three

Hand the whole shortlist to a language model

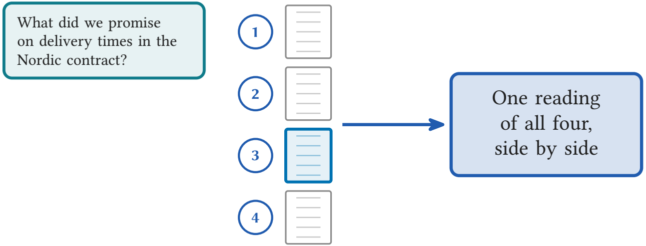
Ask for the order, not for a score



The shortlist, numbered

The model reads the whole shortlist in one prompt and answers with an order, so there is no score left to threshold (RankGPT, 2023).

Ask for the order, not for a score



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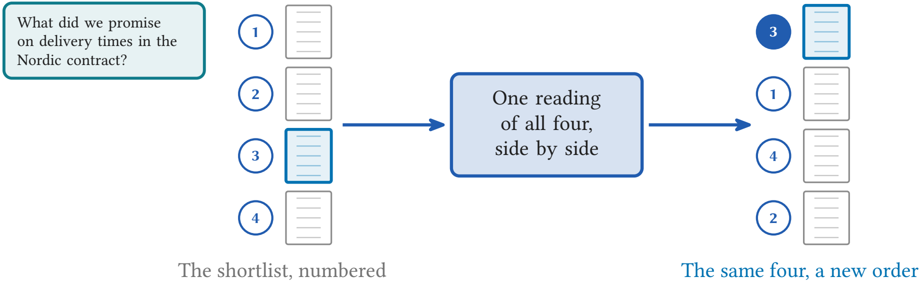

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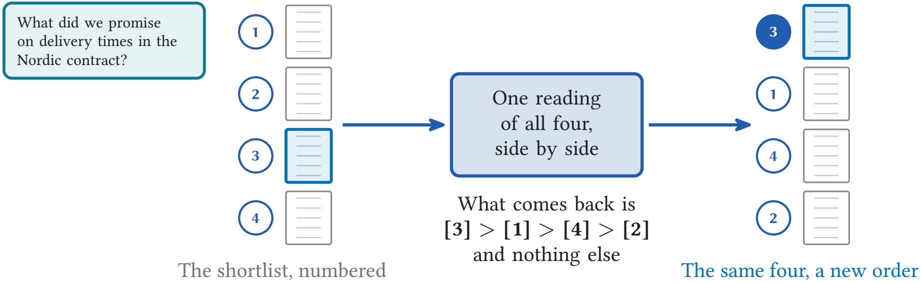


The model reads the whole shortlist in one prompt and answers with an order, so there is no score left to threshold (RankGPT, 2023).

The window slides from the back to the front

What did we promise on delivery times in the Nordic contract?

Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

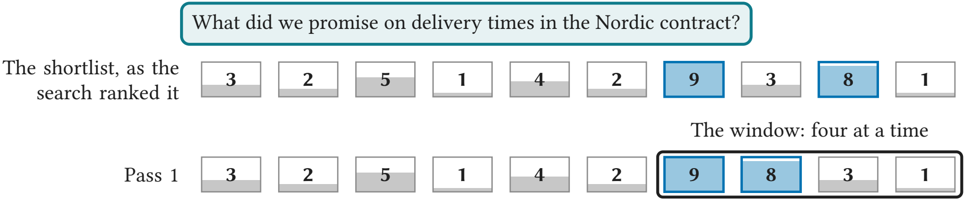
The window slides from the back to the front

What did we promise on delivery times in the Nordic contract?

The shortlist, as the search ranked it	3	2	5	1	4	2	9	3	8	1

Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The window slides from the back to the front



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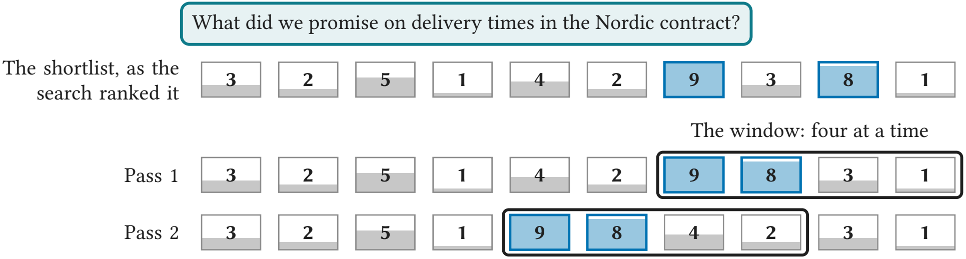

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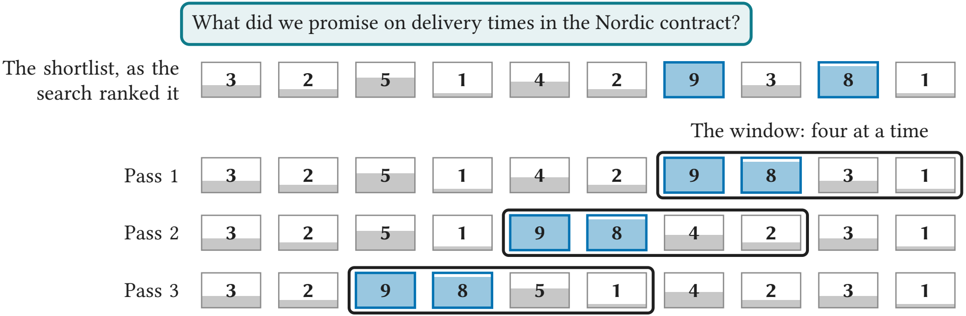

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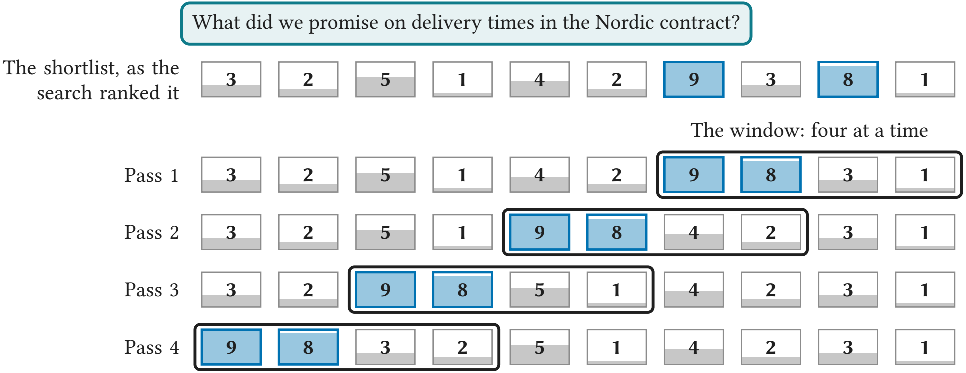

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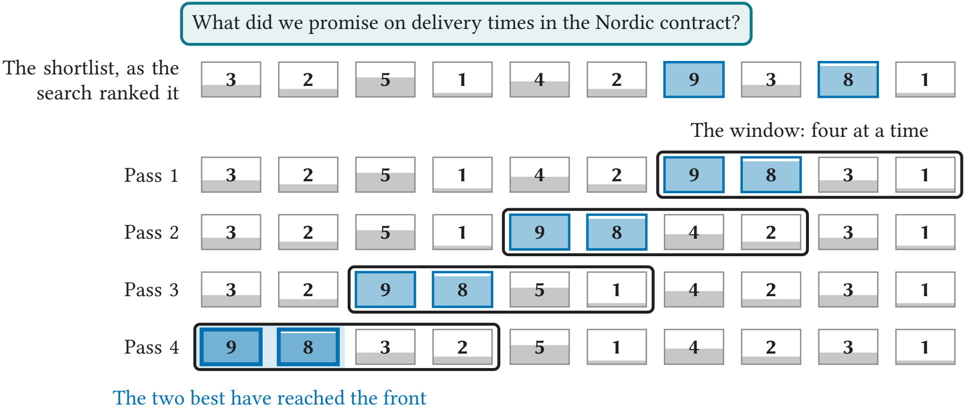

Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The list is longer than the window

The highlighted cell is the page that answers the question

RankGPT: a window of about twenty, stepped ten at a time, so 100 candidates cost ten sequential model calls.

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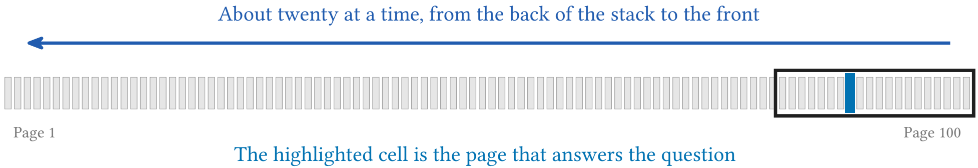

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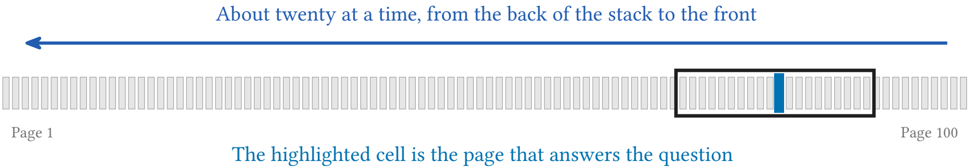

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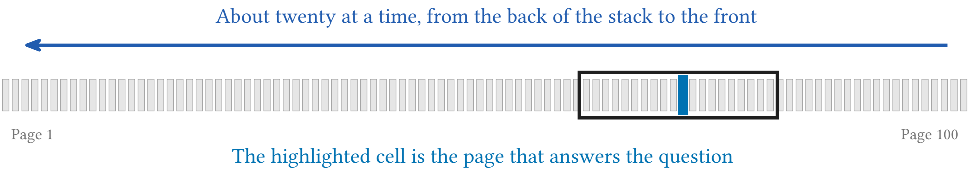
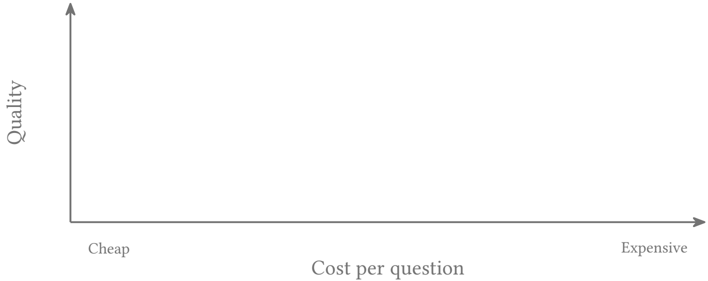


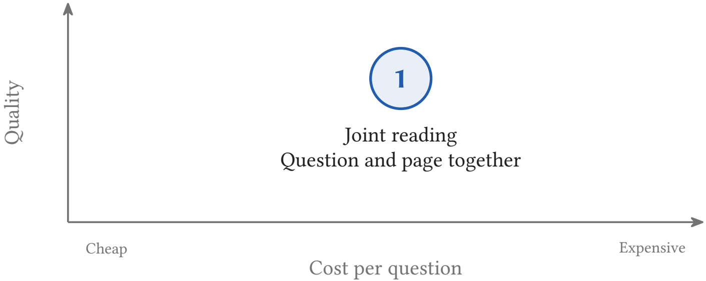
RankGPT: a window of about twenty, stepped ten at a time, so 100 candidates cost ten sequential model calls.

The three ways on one picture



Illustrative, not measured: three papers, three hardware setups, three metrics, no numbers on either axis.

The three ways on one picture



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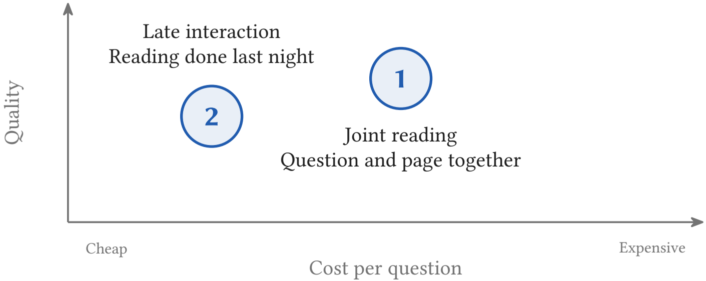
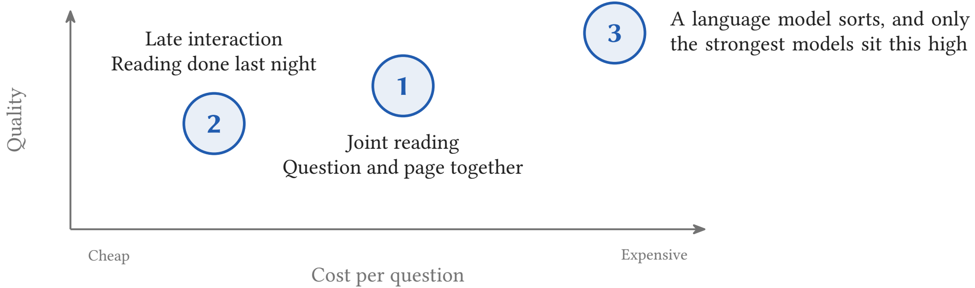
The three ways on one picture



Illustrative, not measured: three papers, three hardware setups, three metrics, no numbers on either axis.

The three ways on one picture



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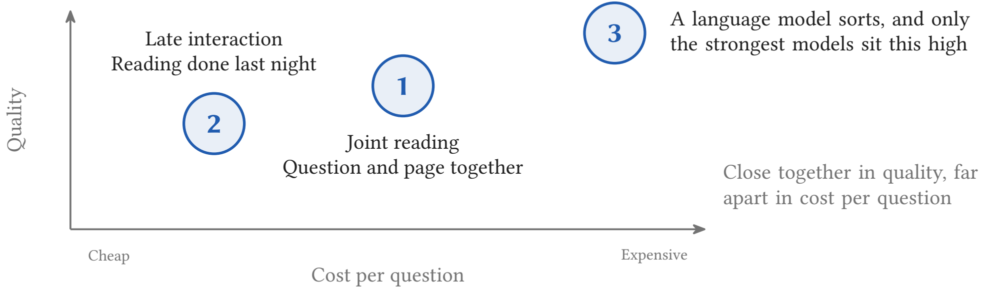
The three ways on one picture



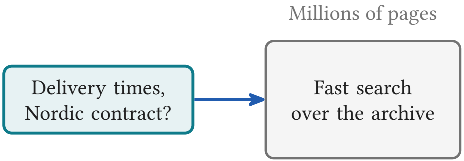
Illustrative, not measured: three papers, three hardware setups, three metrics, no numbers on either axis.

Section 3

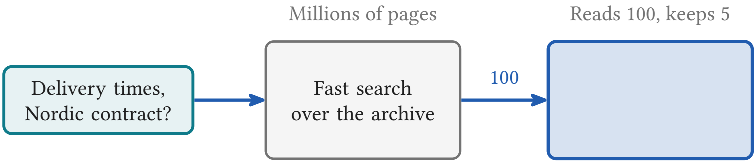
What to remember

3

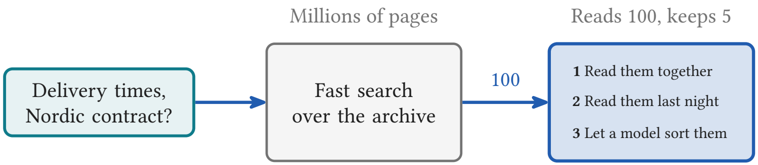
The shape of a real system



The shape of a real system



The shape of a real system



The shape of a real system